APPENDIX A

MOMENTS OF AREAS

A.1. FIRST MOMENT OF AN AREA; CENTROID OF AN AREA

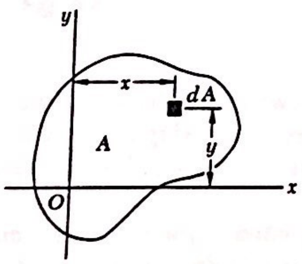


Fig. A.1

Consider an area A located in the xy plane (Fig. A.1). Denoting by x and y the coordinates of an element of area dA, we define the *first moment of the area A with respect to the x axis* as the integral

$$Q_x = \int_A y\, dA \tag{A.1}$$

Similarly, the *first moment of the area A with respect to the y axis* is defined as the integral

$$Q_y = \int_A x\, dA \tag{A.2}$$

We note that each of these integrals may be positive, negative, or zero, depending on the position of the coordinate axes. If SI units are used, the first moments Q_x and Q_y are expressed in m^3 or mm^3; if U.S. customary units are used, they are expressed in ft^3 or in^3.

The *centroid of the area A* is defined as the point C of coordinates $\bar{x}$ and $\bar{y}$ (Fig. A.2), which satisfy the relations

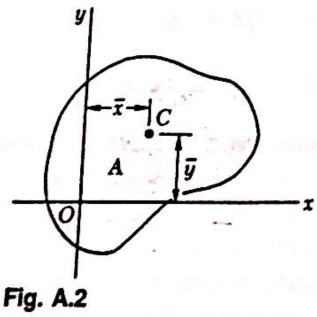


Fig. A.2

$$\int_A x\, dA = A\bar{x} \qquad \int_A y\, dA = A\bar{y} \tag{A.3}$$

Comparing Eqs. (A.1) and (A.2) with Eqs. (A.3), we note that the first moments of the area A may be expressed as the products of the area and of the coordinates of its centroid:

$$Q_x = A\bar{y} \qquad Q_y = A\bar{x} \tag{A.4}$$

When an area possesses an *axis of symmetry*, the first moment of the area with respect to that axis is zero. Indeed, considering the area A of Fig. A.3, which is symmetric with respect to the y axis, we observe that to every element of area dA of abscissa x corresponds an element of area dA' of abscissa $-x$. It follows that the integral in Eq. (A.2) is zero and, thus, that $Q_y = 0$. It also follows from the first of the relations (A.3) that $\bar{x} = 0$. Thus, if an area A possesses an axis of symmetry, its centroid C is located on that axis.

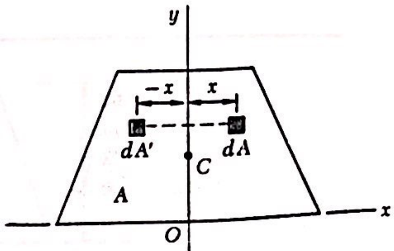
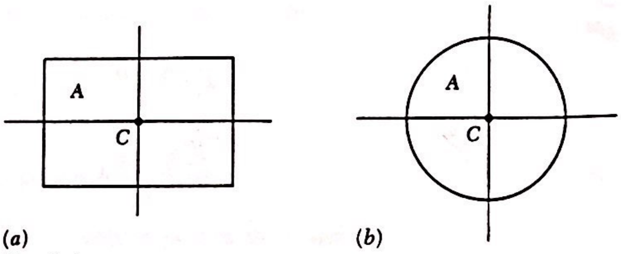


Fig. A.3



(a) (b)
Fig. A.4

Since a rectangle possesses two axes of symmetry (Fig. A.4*a*), the centroid C of a rectangular area coincides with its geometric center. Similarly, the centroid of a circular area coincides with the center of the circle (Fig. A.4*b*).

When an area possesses a *center of symmetry* O, the first moment of the area about any axis through O is zero. Indeed, considering the area A of Fig. A.5, we observe that to every element of area dA of coordinates x and y corresponds an element of area dA' of coordinates $-x$ and $-y$. It follows that the integrals in Eqs. (A.1) and (A.2) are both zero, and that $Q_x = Q_y = 0$. It also follows from Eqs. (A.3) that $\bar{x} = \bar{y} = 0$, that is, the centroid of the area coincides with its center of symmetry.

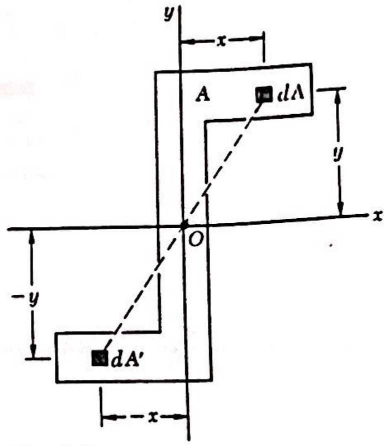


Fig. A.5

When the centroid C of an area can be located by symmetry, the first moment of that area with respect to any given axis may be readily obtained from Eqs. (A.4). For example, in the case of the rectangular area of Fig. A.6, we have

$$Q_x = A\bar{y} = (bh)(\tfrac{1}{2}h) = \tfrac{1}{2}bh^2$$

and

$$Q_y = A\bar{x} = (bh)(\tfrac{1}{2}b) = \tfrac{1}{2}b^2h$$

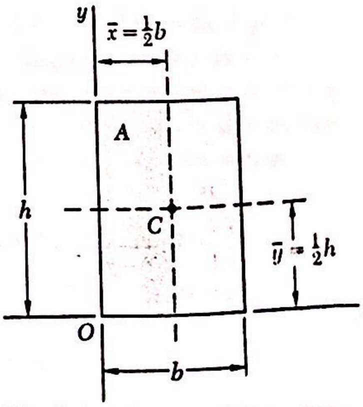


Fig. A.6

In most cases, however, it is necessary to perform the integrations indicated in Eqs. (A.1) through (A.3) to determine the first moments and the centroid of a given area. While each of the integrals involved is actually a double integral, it is possible in many applications to select elements of area dA in the shape of thin horizontal or vertical strips, and thus to reduce the computations to integrations in a single variable. This is illustrated in Example A.01. Centroids of common geometric shapes are indicated in a table inside the back cover of this book.

Example A.01

For the triangular area of Fig. A.7, determine (*a*) the first moment Q_x of the area with respect to the x axis, (*b*) the ordinate $\bar{y}$ of the centroid of the area.

(*a*) ***First Moment*** Q_x. We select as an element of area a horizontal strip of length u and thickness dy, and note that all the points within the element are at the same distance y from the x axis (Fig. A.8). From similar triangles, we have

$$\frac{u}{b} = \frac{h-y}{h} \qquad u = b\frac{h-y}{h}$$

and

$$dA = u\,dy = b\frac{h-y}{h}\,dy$$

The first moment of the area with respect to the x axis is

$$Q_x = \int_A y\,dA = \int_0^h yb\frac{h-y}{h}\,dy = \frac{b}{h}\int_0^h (hy - y^2)\,dy$$

$$= \frac{b}{h}\left[h\frac{y^2}{2} - \frac{y^3}{3}\right]_0^h \qquad Q_x = \tfrac{1}{6}bh^2$$

(*b*) ***Ordinate of Centroid.*** Recalling the first of Eqs. (A.4) and observing that $A = \tfrac{1}{2}bh$, we have

$$Q_x = A\bar{y} \qquad \tfrac{1}{6}bh^2 = (\tfrac{1}{2}bh)\bar{y}$$

$$\bar{y} = \tfrac{1}{3}h$$

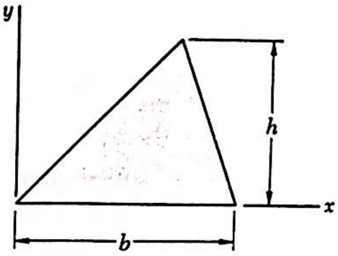


Fig. A.7

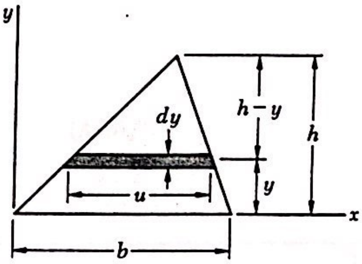


Fig. A.8

areas, and noting that a similar expression may be obtained for Q_y, we write

$$Q_x = \sum_i A_i\bar{y}_i \qquad Q_y = \sum_i A_i\bar{x}_i \tag{A.5}$$

To obtain the coordinates $\overline{X}$ and $\overline{Y}$ of the centroid C of the composite area A, we substitute $Q_x = A\overline{Y}$ and $Q_y = A\overline{X}$ into Eqs. (A.5). We have

$$A\overline{Y} = \sum_i A_i\bar{y}_i \qquad A\overline{X} = \sum_i A_i\bar{x}_i$$

Solving for $\overline{X}$ and $\overline{Y}$ and recalling that the area A is the sum of the component areas A_i, we write

$$\overline{X} = \frac{\sum_i A_i\bar{x}_i}{\sum_i A_i} \qquad \overline{Y} = \frac{\sum_i A_i\bar{y}_i}{\sum_i A_i} \tag{A.6}$$

Example A.02

Locate the centroid C of the area A shown in Fig. A.10.

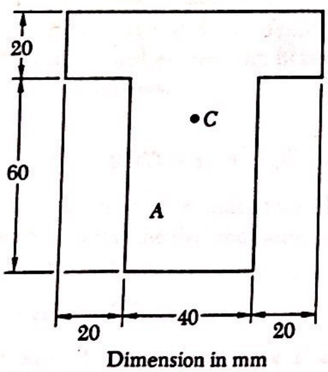


Fig. A.10

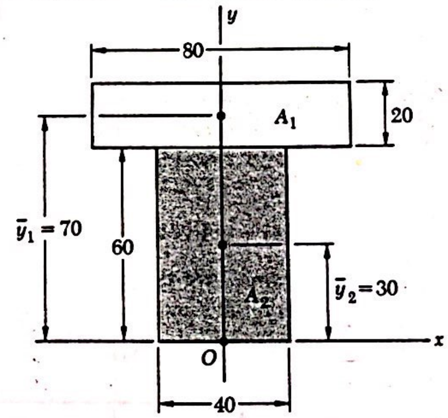


Fig. A.11 Dimensions in mm

Selecting the coordinate axes shown in Fig. A.11, we note that the centroid C must be located on the y axis, since this axis is an axis of symmetry; thus, $\overline{X} = 0$.

	Area, mm²	$\bar{y}_i$, mm	$A_i\bar{y}_i$, mm³
A_1	(20)(80) = 1600	70	112×10^3
A_2	(40)(60) = 2400	30	72×10^3
	$\sum_i A_i = 4000$		$\sum_i A_i\bar{y}_i = 184 \times 10^3$

Dividing A into its component parts A_1 and A_2, we use the second of Eqs. (A.6) to determine the ordinate $\overline{Y}$ of the centroid. The actual computation is best carried out in tabular form.

$$\overline{Y} = \frac{\sum_i A_i\bar{y}_i}{\sum_i A_i} = \frac{184 \times 10^3 \text{ mm}^3}{4 \times 10^3 \text{ mm}^2} = 46 \text{ mm}$$

Example A.03

Referring to the area A of Example A.02, we consider the horizontal x' axis through its centroid C. (Such an axis is called a *centroidal axis*.) Denoting by A' the portion of A located above that axis (Fig. A.12), determine the first moment of A' with respect to the x' axis.

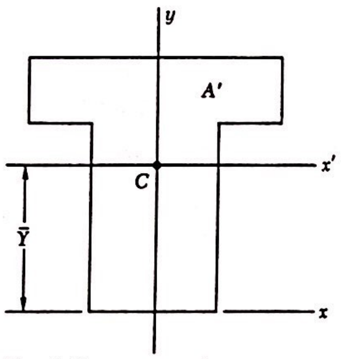


Fig. A.12

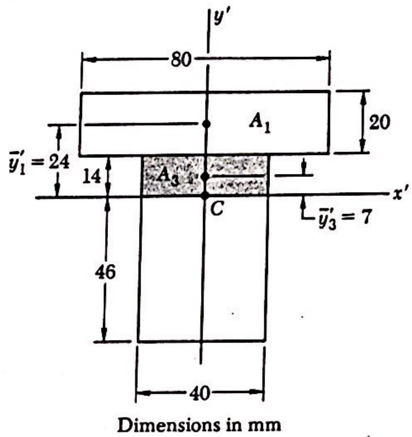


Fig. A.13

Solution. We divide the area A' into its components A_1 and A_3 (Fig. A.13). Recalling from Example A.02 that C is located 46 mm above the lower edge of A, we determine the ordinates $\bar{y}'_1$ and $\bar{y}'_3$ of A_1 and A_3 and express the first moment $Q'_{x'}$ of A' with respect to x' as follows:

$$
\begin{aligned}
Q'_{x'} &= A_1\bar{y}'_1 + A_3\bar{y}'_3 \\
&= (20 \times 80)(24) + (14 \times 40)(7) = 42.3 \times 10^3 \text{ mm}^3
\end{aligned}
$$

Alternative Solution. We first note that since the centroid C of A is located on the x' axis, the first moment $Q_{x'}$ of the *entire area* A with respect to that axis is zero:

$$Q_{x'} = A\bar{y}' = A(0) = 0$$

Denoting by A'' the portion of A located below the x' axis and by $Q''_{x'}$ its first moment with respect to that axis, we have therefore

$$Q_{x'} = Q'_{x'} + Q''_{x'} = 0 \qquad \text{or} \qquad Q'_{x'} = -Q''_{x'}$$

which shows that the first moments of A' and A'' have the same magnitude and opposite signs. Referring to Fig. A.14, we write

$$Q''_{x'} = A_4\bar{y}'_4 = (40 \times 46)(-23) = -42.3 \times 10^3 \text{ mm}^3$$

and

$$Q'_{x'} = -Q''_{x'} = +42.3 \times 10^3 \text{ mm}^3$$

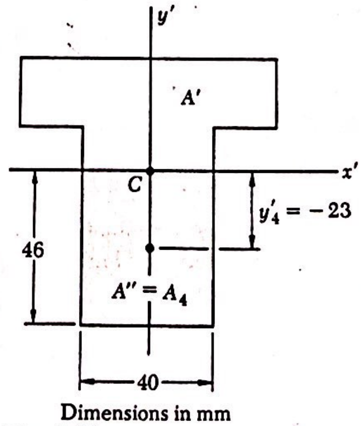


Fig. A.14

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A.3. SECOND MOMENT, OR MOMENT OF INERTIA, OF AN AREA; RADIUS OF GYRATION

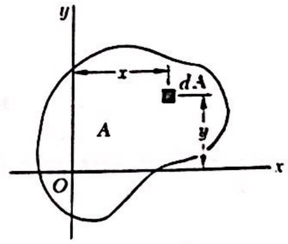


Fig. A.1 (repeated)

Consider again an area A located in the xy plane (Fig. A.1) and the element of area dA of coordinates x and y. The *second moment*, or *moment of inertia*, of the area A with respect to the x axis, and the second moment, or moment of inertia, of A with respect to the y axis are defined, respectively, as

$$I_x = \int_A y^2\,dA \qquad I_y = \int_A x^2\,dA \tag{A.7}$$

These integrals are referred to as *rectangular moments of inertia*, since they are computed from the rectangular coordinates of the element dA. While each integral is actually a double integral, it is possible in many applications to select elements of area dA in the shape of thin horizontal or vertical strips, and thus reduce the computations to integrations in a single variable. This is illustrated in Example A.04.

We now define the *polar moment of inertia* of the area A with respect to point O (Fig. A.15) as the integral

$$J_O = \int_A \rho^2\,dA \tag{A.8}$$

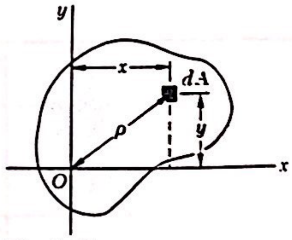


Fig. A.15

where ρ is the distance from O to the element dA. While this integral is again a double integral, it is possible in the case of a circular area to select elements of area dA in the shape of thin circular rings, and thus reduce the computation of J_O to a single integration (see Example A.05).

We note from Eqs. (A.7) and (A.8) that the moments of inertia of an area are positive quantities. If SI units are used, moments of inertia are expressed in m^4 or mm^4.

An important relation may be established between the polar moment of inertia J_O of a given area and the rectangular moments of inertia I_x and I_y of the same area. Noting that $\rho^2 = x^2 + y^2$, we write

$$J_O = \int_A \rho^2\,dA = \int_A (x^2 + y^2)\,dA = \int_A y^2\,dA + \int_A x^2\,dA$$

or

$$J_O = I_x + I_y \tag{A.9}$$

The *radius of gyration* of an area A with respect to the x axis is defined as the quantity r_x, which satisfies the relation

$$I_x = r_x^2 A \tag{A.10}$$

where I_x is the moment of inertia of A with respect to the x axis. Solving Eq. (A.10) for r_x, we have

$$r_x = \sqrt{\frac{I_x}{A}} \tag{A.11}$$

In a similar way, we may define the radii of gyration with respect to the y axis and the origin O. We write

$$I_y = r_y^2 A \qquad r_y = \sqrt{\frac{I_y}{A}} \tag{A.12}$$

$$J_O = r_O^2 A \qquad r_O = \sqrt{\frac{J_O}{A}} \tag{A.13}$$

Substituting for J_O, I_x, and I_y in terms of the corresponding radii of gyration in Eq. (A.9), we observe that

$$r_O^2 = r_x^2 + r_y^2 \tag{A.14}$$

Example A.04

For the rectangular area of Fig. A.16, determine (*a*) the moment of inertia I_x of the area with respect to the centroidal x axis, (*b*) the corresponding radius of gyration r_x.

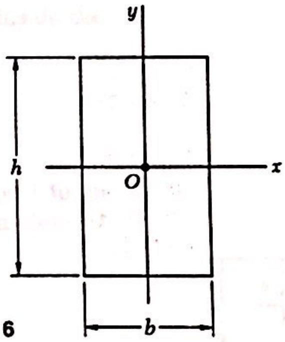


Fig. A.16

(*a*) ***Moment of Inertia*** I_x. We select as an element of area a horizontal strip of length b and thickness dy (Fig. A.17). Since all the points within the strip are at the same distance y from the x axis, the moment of inertia of the strip with respect to that axis is

$$dI_x = y^2\,dA = y^2(b\,dy)$$

Integrating from $y = -h/2$ to $y = +h/2$, we write

$$I_x = \int_A y^2\,dA = \int_{-h/2}^{+h/2} y^2(b\,dy) = \tfrac{1}{3}b[y^3]_{-h/2}^{+h/2}$$

$$= \tfrac{1}{3}b\left(\frac{h^3}{8} + \frac{h^3}{8}\right)$$

or

$$I_x = \tfrac{1}{12}bh^3$$

(*b*) ***Radius of Gyration*** r_x. From Eq. (A.10), we have

$$I_x = r_x^2 A \qquad \tfrac{1}{12}bh^3 = r_x^2(bh)$$

and, solving for r_x,

$$r_x = h/\sqrt{12}$$

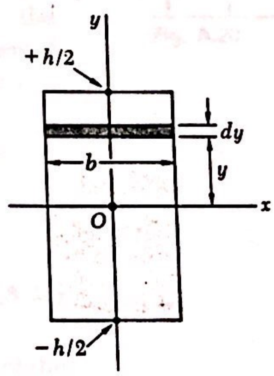


Fig. A.17

Example A.05

For the circular area of Fig. A.18, determine (*a*) the polar moment of inertia J_O, (*b*) the rectangular moments of inertia I_x and I_y.

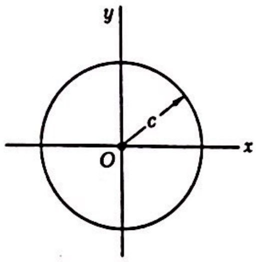


Fig. A.18

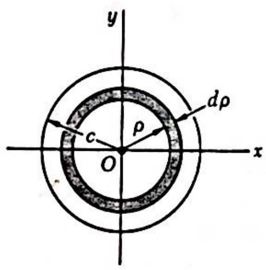


Fig. A.19

(*a*) *Polar Moment of Inertia.* We select as an element of area a ring of radius ρ and thickness $d\rho$ (Fig. A.19). Since all the points within the ring are at the same distance ρ from the origin O, the polar moment of inertia of the ring is

$$dJ_O = \rho^2\, dA = \rho^2(2\pi\rho\, d\rho)$$

Integrating in ρ from 0 to c, we write

$$J_O = \int_A \rho^2\, dA = \int_0^c \rho^2(2\pi\rho\, d\rho) = 2\pi \int_0^c \rho^3\, d\rho$$

$$J_O = \tfrac{1}{2}\pi c^4$$

(*b*) *Rectangular Moments of Inertia.* Because of the symmetry of the circular area, we have $I_x = I_y$. Recalling Eq. (A.9), we write

$$J_O = I_x + I_y = 2I_x \qquad \tfrac{1}{2}\pi c^4 = 2I_x$$

and, thus,

$$I_x = I_y = \tfrac{1}{4}\pi c^4$$

The results obtained in the preceding two examples, and the moments of inertia of other common geometric shapes, are listed in a table inside the back cover of this book.

A.4. PARALLEL-AXIS THEOREM

Consider the moment of inertia I_x of an area A with respect to an arbitrary x axis (Fig. A.20). Denoting by y the distance from an element of area dA to that axis, we recall from Sec. A.3 that

$$I_x = \int_A y^2\, dA$$

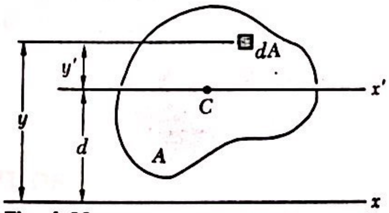


Fig. A.20

Let us now draw the *centroidal x′ axis*, i.e., the axis parallel to the x axis which passes through the centroid C of the area. Denoting by y' the distance from the element dA to that axis, we write $y = y' + d$, where d is the distance between the two axes. Substituting for y in the integral representing I_x, we write

$$I_x = \int_A y^2 dA = \int_A (y' + d)^2 dA$$

$$I_x = \int_A y'^2\, dA + 2d \int_A y'\, dA + d^2 \int_A dA \qquad \text{(A.15)}$$

The first integral in Eq. (A.15) represents the moment of inertia $\bar{I}_{x'}$ of the area with respect to the centroidal x' axis. The second integral represents the first moment $Q_{x'}$ of the area with respect to the x' axis and is equal to zero, since the centroid C of the area is located on that axis. Indeed, we recall from Sec. A.1 that

$$Q_{x'} = A\bar{y}' = A(0) = 0$$

Finally, we observe that the last integral in Eq. (A.15) is equal to the total area A. We have, therefore,

$$\boxed{I_x = \bar{I}_{x'} + Ad^2} \tag{A.16}$$

This formula expresses that the moment of inertia I_x of an area with respect to an arbitrary x axis is equal to the moment of inertia $\bar{I}_{x'}$ of the area with respect to the centroidal x' axis parallel to the x axis, *plus* the product Ad^2 of the area A and of the square of the distance d between the two axes. This result is known as the *parallel-axis theorem.* It makes it possible to determine the moment of inertia of an area with respect to a given axis, when its moment of inertia with respect to a centroidal axis of the same direction is known. Conversely, it makes it possible to determine the moment of inertia $\bar{I}_{x'}$ of an area A with respect to a centroidal axis x', when the moment of inertia I_x of A with respect to a parallel axis is known, by *subtracting* from I_x the product Ad^2. We should note that the parallel-axis theorem may be used *only if one of the two axes involved is a centroidal axis.*

A similar formula may be derived, which relates the polar moment of inertia J_O of an area with respect to an arbitrary point O and the polar moment of inertia $\bar{J}_C$ of the same area with respect to its centroid C. Denoting by d the distance between O and C, we write

$$J_O = \bar{J}_C + Ad^2 \tag{A.17}$$

A.5. DETERMINATION OF THE MOMENT OF INERTIA OF A COMPOSITE AREA

Consider a composite area A made of several component parts A_1, A_2, and so forth. Since the integral representing the moment of inertia of A may be subdivided into integrals extending over A_1, A_2, and so forth, the moment of inertia of A with respect to a given axis will be obtained by adding the moments of inertia of the areas A_1, A_2, and so forth, with respect to the same axis. The moment of inertia of an area made of several of the common shapes shown in the table inside the back cover of this book may thus be obtained from the formulas given in that table. Before adding the moments of inertia of the component areas, however, the parallel-axis theorem should be used to transfer each moment of inertia to the desired axis. This is shown in Example A.06.

Example A.06

Determine the moment of inertia $\bar{I}_x$ of the area shown with respect to the centroidal x axis (Fig. A.21).

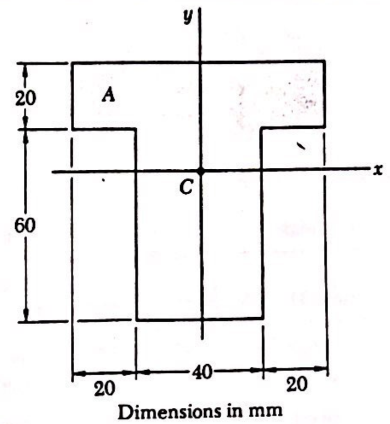


Fig. A.21

Location of Centroid. The centroid C of the area must first be located. However, this has already been done in Example A.02 for the given area. We recall from that example that C is located 46 mm above the lower edge of the area A.

Computation of Moment of Inertia. We divide the area A into the two rectangular areas A_1 and A_2 (Fig. A.22), and compute the moment of inertia of each area with respect to the x axis.

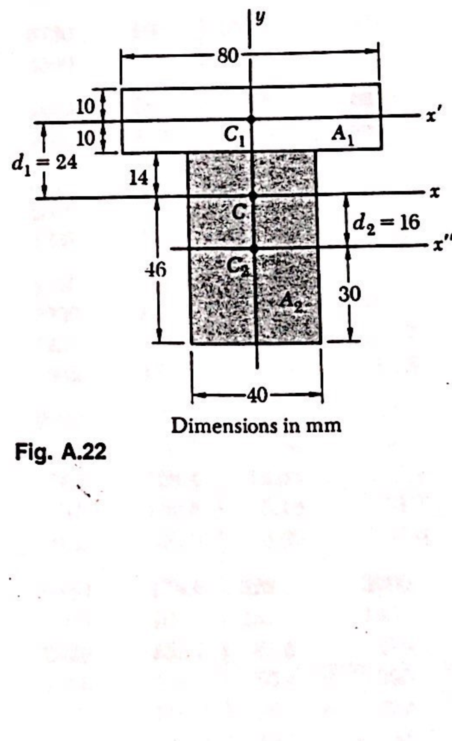


Fig. A.22

Rectangular Area A_1. To obtain the moment of inertia $(I_x)_1$ of A_1 with respect to the x axis, we first compute the moment of inertia of A_1 *with respect to its own centroidal axis x'*. Recalling the formula derived in part *a* of Example A.04 for the centroidal moment of inertia of a rectangular area, we have

$$(\bar{I}_{x'})_1 = \tfrac{1}{12}bh^3 = \tfrac{1}{12}(80\text{ mm})(20\text{ mm})^3 = 53.3 \times 10^3\text{ mm}^4$$

Using the parallel-axis theorem, we transfer the moment of inertia of A_1 from its centroidal axis x' to the parallel axis x:

$$\begin{aligned}(I_x)_1 &= (\bar{I}_{x'})_1 + A_1 d_1^2 = 53.3 \times 10^3 + (80 \times 20)(24)^2\\ &= 975 \times 10^3\text{ mm}^4\end{aligned}$$

Rectangular Area A_2. Computing the moment of inertia of A_2 with respect to its centroidal axis x'', and using the parallel-axis theorem to transfer it to the x axis, we have

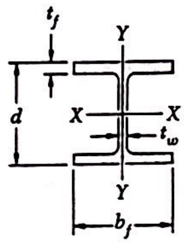
$$(\bar{I}_{x''})_2 = \tfrac{1}{12}bh^3 = \tfrac{1}{12}(40)(60)^3 = 720 \times 10^3\text{ mm}^4$$

$$\begin{aligned}(I_x)_2 &= (\bar{I}_{x''})_2 + A_2 d_2^2 = 720 \times 10^3 + (40 \times 60)(16)^2\\ &= 1334 \times 10^3\text{ mm}^4\end{aligned}$$

Entire Area A. Adding the values computed for the moments of inertia of A_1 and A_2 with respect to the x axis, we obtain the moment of inertia $\bar{I}_x$ of the entire area:

$$\bar{I}_x = (I_x)_1 + (I_x)_2 = 975 \times 10^3 + 1334 \times 10^3$$

$$\bar{I}_x = 2.31 \times 10^6\text{ mm}^4$$



Appendix C. Properties of Rolled-Steel Shapes
(SI Units)

W Shapes
(Wide-Flange Shapes)

			Flange		Web	Axis X-X			Axis Y-Y		
Designation†	Area A, mm²	Depth d, mm	Width b_f, mm	Thickness t_f, mm	Web Thickness t_w, mm	I_x 10^6 mm⁴	S_x 10^3 mm³	r_x mm	I_y 10^6 mm⁴	S_y 10^3 mm³	r_y mm
W920 × 446	57000	933	423	42.7	24.0	8450	18110	386	541	2560	97.3
201	25600	903	304	20.1	15.2	3250	7200	356	93.7	616	60.5
W840 × 299	38100	855	400	29.2	18.2	4790	11200	356	312	1560	90.4
176	22400	835	292	18.8	14.0	2460	5890	330	77.8	533	58.9
W760 × 257	32800	773	381	27.1	16.6	3410	8820	323	249	1307	87.1
147	18800	753	265	17.0	13.2	1660	4410	297	53.3	402	53.3
W690 × 217	27700	695	355	24.8	15.4	2340	6730	290	184.4	1039	81.5
125	16000	678	253	16.3	11.7	1186	3500	272	44.1	349	52.6
W610 × 155	19700	611	324	19.0	12.7	1290	4220	256	107.8	665	73.9
101	13000	603	228	14.9	10.5	762	2530	243	29.3	257	47.5
W530 × 150	19200	543	312	20.3	12.7	1007	3710	229	103.2	662	73.4
92	11800	533	209	15.6	10.2	554	2080	217	23.9	229	45.0
66	8390	525	165	11.4	8.9	351	1337	205	8.62	104.5	32.0
W460 × 158	20100	476	284	23.9	15.0	795	3340	199.1	91.6	645	67.6
113	14400	463	280	17.3	10.8	554	2390	196.3	63.3	452	66.3
74	9480	457	190	14.5	9.0	333	1457	187.5	16.69	175.7	41.9
52	6650	450	152	10.8	7.6	212	942	178.8	6.37	83.8	31.0
W410 × 114	14600	420	261	19.3	11.6	462	2200	177.8	57.4	440	62.7
85	10800	417	181	18.2	10.9	316	1516	170.7	17.94	198.2	40.6
60	7610	407	178	12.8	7.7	216	1061	168.4	12.03	135.2	39.9
46.1	5880	403	140	11.2	7.0	156.1	775	162.8	5.16	73.7	29.7
38.8	4950	399	140	8.8	6.4	125.3	628	159.0	3.99	57.0	28.4
W360 × 551	70300	455	418	67.6	42.0	2260	9930	179.6	828	3960	108.5
216	27500	375	394	27.7	17.3	712	3800	160.8	282	1431	101.1
122	15500	363	257	21.7	13.0	367	2020	153.7	61.6	479	63.0
101	12900	357	255	18.3	10.5	301	1686	152.7	50.4	395	62.5
79	10100	354	205	16.8	9.4	225	1271	149.6	24.0	234	48.8
64	8130	347	203	13.5	7.7	178.1	1027	147.8	18.81	185.3	48.0
57	7230	358	172	13.1	7.9	160.2	895	149.4	11.11	129.2	39.4
44.8	5710	352	171	9.8	6.9	121.1	688	145.5	8.16	95.4	37.8
39.0	4960	353	128	10.7	6.5	102.0	578	143.5	3.71	58.0	27.4
32.9	4190	349	127	8.5	5.8	82.8	474	140.7	2.91	45.8	26.4

†A wide-flange shape is designated by the letter W followed by the nominal depth in millimeters and the mass in kilograms per meter.

(Table continued on page 707)

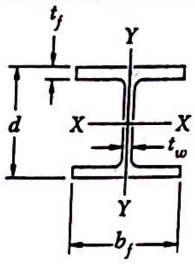
Appendix C. Properties of Rolled-Steel Shapes

(SI Units)

Continued from page 705

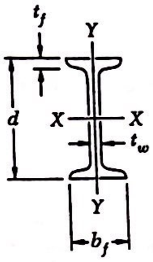
W Shapes

(Wide-Flange Shapes)



Designation†	Area A, mm²	Depth d, mm	Flange: Width b_f, mm	Flange: Thickness t_f, mm	Web Thickness t_w, mm	Axis X-X: I_x 10^6 mm⁴	Axis X-X: S_x 10^3 mm³	Axis X-X: r_x mm	Axis Y-Y: I_y 10^6 mm⁴	Axis Y-Y: S_y 10^3 mm³	Axis Y-Y: r_y mm
W310 × 143	18200	323	309	22.9	14.0	347	2150	138.2	112.4	728	78.5
107	13600	311	306	17.0	10.9	248	1595	134.9	81.2	531	77.2
74	9480	310	205	16.3	9.4	164.0	1058	131.6	23.4	228	49.8
60	7610	303	203	13.1	7.5	129.0	851	130.3	18.36	180.9	49.0
52	6650	317	167	13.2	7.6	118.6	748	133.4	10.20	122.2	39.1
44.5	5670	313	166	11.2	6.6	99.1	633	132.3	8.45	101.8	38.6
38.7	4940	310	165	9.7	5.8	84.9	548	131.3	7.20	87.3	38.4
32.7	4180	313	102	10.8	6.6	64.9	415	124.7	1.940	38.0	21.5
23.8	3040	305	101	6.7	5.6	42.9	281	118.6	1.174	23.2	19.63
W250 × 167	21200	289	265	31.8	19.2	298.0	2060	118.4	98.2	741	68.1
101	12900	264	257	19.6	11.9	164.0	1242	112.8	55.8	434	65.8
80	10200	256	255	15.6	9.4	126.1	985	111.0	42.8	336	65.0
67	8580	257	204	15.7	8.9	103.2	803	110.0	22.2	218	51.1
58	7420	252	203	13.5	8.0	87.0	690	108.5	18.73	184.5	50.3
49.1	6260	247	202	11.0	7.4	70.8	573	106.4	15.23	150.8	49.3
44.8	5700	266	148	13.0	7.6	70.8	532	111.3	6.95	93.9	34.8
32.7	4190	258	146	9.1	6.1	49.1	381	108.5	4.75	65.1	33.8
28.4	3630	260	102	10.0	6.4	40.1	308	105.2	1.796	35.2	22.2
22.3	2850	254	102	6.9	5.8	28.7	226	100.3	1.203	23.6	20.6
W200 × 86	11000	222	209	20.6	13.0	94.9	855	92.7	31.3	300	53.3
71	9100	216	206	17.4	10.2	76.6	709	91.7	25.3	246	52.8
59	7550	210	205	14.2	9.1	60.8	579	89.7	20.4	199.0	51.8
52	6650	206	204	12.6	7.9	52.9	514	89.2	17.73	173.8	51.6
46.1	5890	203	203	11.0	7.2	45.8	451	88.1	15.44	152.1	51.3
41.7	5320	205	166	11.8	7.2	40.8	398	87.6	9.03	108.8	41.1
35.9	4570	201	165	10.2	6.2	34.5	343	86.9	7.62	92.4	40.9
31.3	3970	210	134	10.2	6.4	31.3	298	88.6	4.07	60.7	32.0
26.6	3390	207	133	8.4	5.8	25.8	249	87.1	3.32	49.9	31.2
22.5	2860	206	102	8.0	6.2	20.0	194.2	83.6	1.419	27.8	22.3
19.3	2480	203	102	6.5	5.8	16.48	162.4	81.5	1.136	22.3	21.4
W150 × 37.1	4740	162	154	11.6	8.1	22.2	274	68.6	7.12	92.5	38.6
29.8	3790	157	153	9.3	6.6	17.23	219	67.6	5.54	72.4	38.1
24.0	3060	160	102	10.3	6.6	13.36	167.0	66.0	1.844	36.2	24.6
18.0	2290	153	102	7.1	5.8	9.20	120.3	63.2	1.245	24.4	23.3
13.5	1730	150	100	5.5	4.3	6.83	91.1	62.7	0.916	18.32	23.0
W130 × 28.1	3590	131	128	10.9	6.9	10.91	166.6	55.1	3.80	59.4	32.5
23.8	3040	127	127	9.1	6.1	8.87	139.7	54.1	3.13	49.3	32.3
W100 × 19.3	2470	106	103	8.8	7.1	4.70	88.7	43.7	1.607	31.2	25.4

†A wide-flange shape is designated by the letter W followed by the nominal depth in millimeters and the mass in kilograms per meter.



Appendix C. **Properties of Rolled-Steel Shapes**
(SI Units)

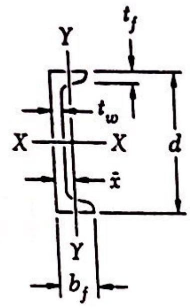
S Shapes
(American Standard Shapes)

Designation†	Area A, mm²	Depth d, mm	Flange Width b_f, mm	Flange Thickness t_f, mm	Web Thickness t_w, mm	Axis X-X I_x 10^6 mm⁴	Axis X-X S_x 10^3 mm³	Axis X-X r_x mm	Axis Y-Y I_y 10^6 mm⁴	Axis Y-Y S_y 10^3 mm³	Axis Y-Y r_y mm
S610 × 149	18970	610	184	22.1	19.0	995	3260	229	19.90	216	32.3
134	17100	610	181	22.1	15.8	937	3070	234	18.69	207	33.0
118.9	15160	610	178	22.1	12.7	878	2880	241	17.61	197.9	34.0
S510 × 141	18000	508	183	23.3	20.3	670	2640	193.0	20.69	226.	33.8
127	16130	508	179	23.3	16.6	633	2490	197.9	19.23	215.	34.5
112	14260	508	162	20.1	16.3	533	2100	193.0	12.32	152.1	29.5
97.3	12390	508	159	20.1	12.7	491	1933	199.1	11.40	143.4	30.2
S460 × 104	13290	457	159	17.6	18.1	385	1685	170.4	10.03	126.2	27.4
81.4	10390	457	152	17.6	11.7	335	1466	179.6	8.66	113.9	29.0
S380 × 74	9480	381	143	15.8	14.0	202	1060	146.1	6.53	91.3	26.2
64	8130	381	140	15.8	10.4	186.1	977	151.1	5.99	85.6	27.2
S310 × 74	9480	305	139	16.8	17.4	127.0	833	115.6	6.53	94.0	26.2
60.7	7740	305	133	16.8	11.7	113.2	742	121.2	5.66	85.1	26.9
52	6640	305	129	13.8	10.9	95.3	625	119.9	4.11	63.7	24.9
47.3	6032	305	127	13.8	8.9	90.7	595	122.7	3.90	61.4	25.4
S250 × 52	6640	254	126	12.5	15.1	61.2	482	96.0	3.48	55.2	22.9
37.8	4806	254	118	12.5	7.9	51.6	406	103.4	2.83	48.0	24.2
S200 × 34	4368	203	106	10.8	11.2	27.0	266	78.7	1.794	33.8	20.3
27.4	3484	203	102	10.8	6.9	24.0	236	82.8	1.553	30.4	21.1
S180 × 30	3794	178	97	10.0	11.4	17.65	198.3	68.3	1.319	27.2	18.64
22.8	2890	178	92	10.0	6.4	15.28	171.7	72.6	1.099	23.9	19.45
S150 × 25.7	3271	152	90	9.1	11.8	10.95	144.1	57.9	0.961	21.4	17.15
18.6	2362	152	84	9.1	5.8	9.20	121.1	62.2	0.758	18.05	17.91
S130 × 22.0	2800	127	83	8.3	12.5	6.33	99.7	47.5	0.695	16.75	15.75
15	1884	127	76	8.3	5.3	5.12	80.6	52.1	0.508	13.37	16.33
S100 × 14.1	1800	102	70	7.4	8.3	2.83	55.5	39.6	0.376	10.74	14.45
11.5	1452	102	67	7.4	4.8	2.53	49.6	41.6	0.318	9.49	14.75
S75 × 11.2	1426	76	63	6.6	8.9	1.22	32.1	29.2	0.244	7.75	13.11
8.5	1077	76	59	6.6	4.3	1.05	27.6	31.3	0.189	6.41	13.26

†An American Standard Beam is designated by the letter S followed by the nominal depth in millimeters and the mass in kilograms per meter.

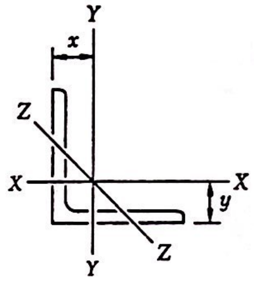
Appendix C. Properties of Rolled-Steel Shapes
(SI Units)

C Shapes
(American Standard Channels)



Designation†	Area A, mm²	Depth d, mm	Flange Width b_f, mm	Flange Thickness t_f, mm	Web Thickness t_w, mm	Axis X-X I_x 10^6 mm⁴	Axis X-X S_x 10^3 mm³	Axis X-X r_x mm	Axis Y-Y I_y 10^6 mm⁴	Axis Y-Y S_y 10^3 mm³	Axis Y-Y r_y mm	$\bar{x}$ mm
C380 × 74	9480	381	94	16.5	18.2	168.2	883	133.1	4.58	62.1	22.0	20.3
60	7610	381	89	16.5	13.2	145.3	763	138.2	3.84	55.5	22.5	19.76
50.4	6426	381	86	16.5	10.2	131.1	688	142.7	3.38	51.2	23.0	19.99
C310 × 45	5690	305	80	12.7	13.0	67.4	442	109.0	2.14	34.0	19.38	17.12
37	4742	305	77	12.7	9.8	59.9	393	112.5	1.861	31.1	19.81	17.12
30.8	3929	305	74	12.7	7.2	53.7	352	117.1	1.615	28.7	20.29	17.73
C250 × 45	5690	254	76	11.1	17.1	42.9	338	86.9	1.640	27.6	16.99	16.48
37	4742	254	73	11.1	13.4	38.0	299	89.4	1.399	24.4	17.17	15.67
30	3794	254	69	11.1	9.6	32.8	258	93.0	1.170	21.8	17.55	15.39
22.8	2897	254	65	11.1	6.1	28.1	221	98.3	0.949	18.29	18.11	16.10
C230 × 30	3794	229	67	10.5	11.4	25.4	222	81.8	1.007	19.29	16.31	14.81
22	2845	229	63	10.5	7.2	21.2	185.2	86.4	0.803	16.69	16.79	14.88
19.9	2542	229	61	10.5	5.9	19.94	174.2	88.4	0.733	16.03	16.97	15.27
C200 × 27.9	3555	203	64	9.9	12.4	18.31	180.4	71.6	0.824	16.60	15.21	14.35
20.5	2606	203	59	9.9	7.7	15.03	148.1	75.9	0.637	14.17	15.62	14.05
17.1	2181	203	57	9.9	5.6	13.57	133.7	79.0	0.549	12.92	15.88	14.50
C180 × 22.0	2794	178	58	9.3	10.6	11.32	127.2	63.8	0.574	12.90	14.33	13.51
18.2	2323	178	55	9.3	8.0	10.07	113.2	66.0	0.487	11.69	14.50	13.34
14.6	1852	178	53	9.3	5.3	8.86	99.6	69.1	0.403	10.26	14.76	13.74
C150 × 19.3	2471	152	54	8.7	11.1	7.24	95.3	54.1	0.437	10.67	13.34	13.06
15.6	1994	152	51	8.7	8.0	6.33	83.3	56.4	0.360	9.40	13.44	12.70
12.2	1548	152	48	8.7	5.1	5.45	71.7	59.4	0.288	8.23	13.64	13.00
C130 × 13.4	1703	127	47	8.1	8.3	3.70	58.3	46.5	0.263	7.54	12.42	12.14
10.0	1271	127	44	8.1	4.8	3.12	49.1	49.5	0.199	6.28	12.52	12.29
C100 × 10.8	1374	102	43	7.5	8.2	1.911	37.5	37.3	0.180	5.74	11.43	11.66
8.0	1026	102	40	7.5	4.7	1.602	31.4	39.6	0.133	4.69	11.40	11.63
C75 × 8.9	1135	76	40	6.9	9.0	0.862	22.7	27.4	0.127	4.47	10.57	11.56
7.4	948	76	37	6.9	6.6	0.770	20.3	28.4	0.103	3.98	10.41	11.13
6.1	781	76	35	6.9	4.3	0.691	18.18	29.7	0.082	3.43	10.26	11.10

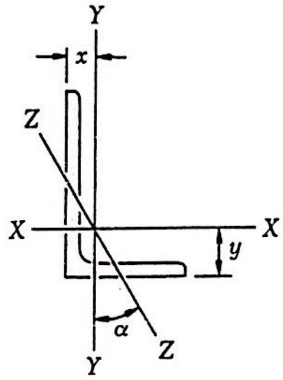
†An American Standard Channel is designated by the letter C followed by the nominal depth in millimeters and the mass in kilograms per meter



Appendix C. **Properties of Rolled-Steel Shapes**
(SI Units)

Angles
Equal Legs

Size and Thickness, mm	Mass per Meter, kg/m	Area, mm²	Axis X-X and Axis Y-Y: I 10^6 mm^4	S 10^3 mm^3	r mm	x or y mm	Axis Z-Z: r mm
L203 × 203 × 25.4	75.9	9680	37.0	259	61.8	60.2	39.6
19.0	57.9	7360	29.0	200	62.8	57.9	40.1
12.7	39.3	5000	20.2	137.0	63.6	55.6	40.4
L152 × 152 × 25.4	55.7	7100	14.78	140.4	45.6	47.2	29.7
19.0	42.7	5445	11.74	109.1	46.4	45.2	29.7
15.9	36.0	4590	10.07	92.8	46.8	43.9	30.0
12.7	29.2	3710	8.28	75.5	47.2	42.7	30.0
9.5	22.2	2800	6.41	57.8	47.8	41.7	30.2
L127 × 127 × 19.0	35.1	4480	6.53	74.2	38.2	38.6	24.8
15.9	29.8	3780	5.66	63.3	38.7	37.6	24.8
12.7	24.1	3070	4.70	51.8	39.2	36.3	25.0
9.5	18.3	2330	3.64	39.7	39.5	35.3	25.1
L102 × 102 × 19.0	27.5	3510	3.19	46.0	30.1	32.3	19.76
15.9	23.4	2970	2.77	39.3	30.5	31.2	19.79
12.7	19.0	2420	2.31	32.3	30.9	30.0	19.86
9.5	14.6	1845	1.815	24.9	31.4	29.0	20.0
6.4	9.8	1252	1.265	17.21	31.8	27.7	20.2
L89 × 89 × 12.7	16.5	2100	1.515	24.4	26.9	26.9	17.35
9.5	12.6	1600	1.195	18.85	27.3	25.7	17.45
6.4	8.6	1090	0.837	13.01	27.7	24.6	17.63
L76 × 76 × 12.7	14.0	1774	0.924	17.53	22.8	23.7	14.83
9.5	10.7	1361	0.733	13.65	23.2	22.6	14.91
6.4	7.3	929	0.516	9.46	23.6	21.4	15.04
L64 × 64 × 12.7	11.4	1452	0.512	11.86	18.78	20.5	12.37
9.5	8.7	1116	0.410	9.28	19.17	19.35	12.37
6.4	6.1	768	0.293	6.46	19.53	18.21	12.47
4.8	4.6	581	0.228	4.97	19.81	17.63	12.57
L51 × 51 × 9.5	7.0	877	0.1994	5.75	15.08	16.15	9.88
6.4	4.7	605	0.1448	4.05	15.47	15.04	9.93
3.2	2.4	312	0.0791	2.15	15.92	13.87	10.11



Appendix C. Properties of Rolled-Steel Shapes
(SI Units)

Angles
Unequal Legs

			Axis X-X				Axis Y-Y				Axis Z-Z	
Size and Thickness, mm	Mass per Meter kg/m	Area mm^2	I_x 10^6 mm^4	S_x 10^3 mm^3	r_x mm	y mm	I_y 10^6 mm^4	S_y 10^3 mm^3	r_y mm	x mm	r_z mm	tan α
L203 × 152 × 25.4	65.5	8390	33.6	247	63.3	67.3	16.15	146.2	43.9	41.9	32.5	0.543
19.0	50.1	6410	26.4	192	64.2	65.0	12.78	113.4	44.7	39.6	32.8	0.551
12.7	34.1	4350	18.44	131	65.1	62.7	9.03	78.5	45.6	37.3	33.0	0.558
L152 × 102 × 19.0	35.0	4480	10.20	102.4	47.7	52.8	3.61	48.7	28.4	27.4	21.8	0.428
12.7	24.0	3060	7.24	71.0	48.6	50.5	2.61	34.1	29.2	25.1	22.1	0.440
9.5	18.2	2330	5.62	54.4	49.1	49.3	2.04	26.2	29.6	23.9	22.3	0.446
L127 × 76 × 12.7	19.0	2420	3.93	47.7	40.3	44.5	1.074	18.85	21.1	19.05	16.46	0.357
9.5	14.5	1845	3.07	36.7	40.8	43.2	0.849	14.55	21.5	17.88	16.61	0.364
6.4	9.8	1252	2.13	25.1	41.2	42.2	0.599	10.06	21.9	16.69	16.84	0.371
L102 × 76 × 12.7	16.4	2100	2.10	31.0	31.6	33.8	1.007	18.35	21.9	21.0	16.23	0.543
9.5	12.6	1600	1.648	23.9	32.1	32.5	0.799	14.19	22.3	19.86	16.36	0.551
6.4	8.6	1090	1.153	16.39	32.5	31.5	0.566	9.82	22.8	18.69	16.54	0.558
L89 × 64 × 12.7	13.9	1774	1.349	23.1	27.6	30.5	0.566	12.45	17.88	17.91	13.56	0.486
9.5	10.7	1361	1.066	17.86	28.0	29.5	0.454	9.70	18.26	16.76	13.64	0.496
6.4	7.3	929	0.749	12.37	28.4	28.2	0.323	6.75	18.65	15.60	13.82	0.506
L76 × 51 × 12.7	11.5	1452	0.799	16.39	23.5	27.4	0.280	7.77	13.89	14.81	10.87	0.414
9.5	8.8	1116	0.637	12.80	23.9	26.4	0.226	6.08	14.20	13.69	10.92	0.428
6.4	6.1	768	0.454	8.88	24.3	25.2	0.1632	4.26	14.58	12.52	11.05	0.440
L64 × 51 × 9.5	7.9	1000	0.380	8.96	19.51	21.1	0.214	5.95	14.66	14.76	10.67	0.614
6.4	5.4	684	0.272	6.24	19.94	20.0	0.1548	4.16	15.04	13.64	10.77	0.626